Northeastern University

Teleoperation of Semi-Humanoid for Construction with Underactuated Dexterous Hand

Xiao Hu[1], Yangming Shi[2], Yang Ye[3*]

[1] Ph.D. Student, Informatics, Department of Civil and Environmental Engineering, Northeastern University, Boston, USA;
Email: xiao.h1@northeastern.edu;

[2] Assistant Professor, Department of Civil and Environmental Engineering, Colorado School of Mines, USA;
Email: yangming.shi@mines.edu;

[3] Assistant Professor, Informatics, Department of Civil and Environmental Engineering, Northeastern University, Boston, USA;
Email: y.ye@northeastern.edu; Corresponding Author.

Background:
Robot Teleoperation



Operation safety
(e.g., asbestos removal)

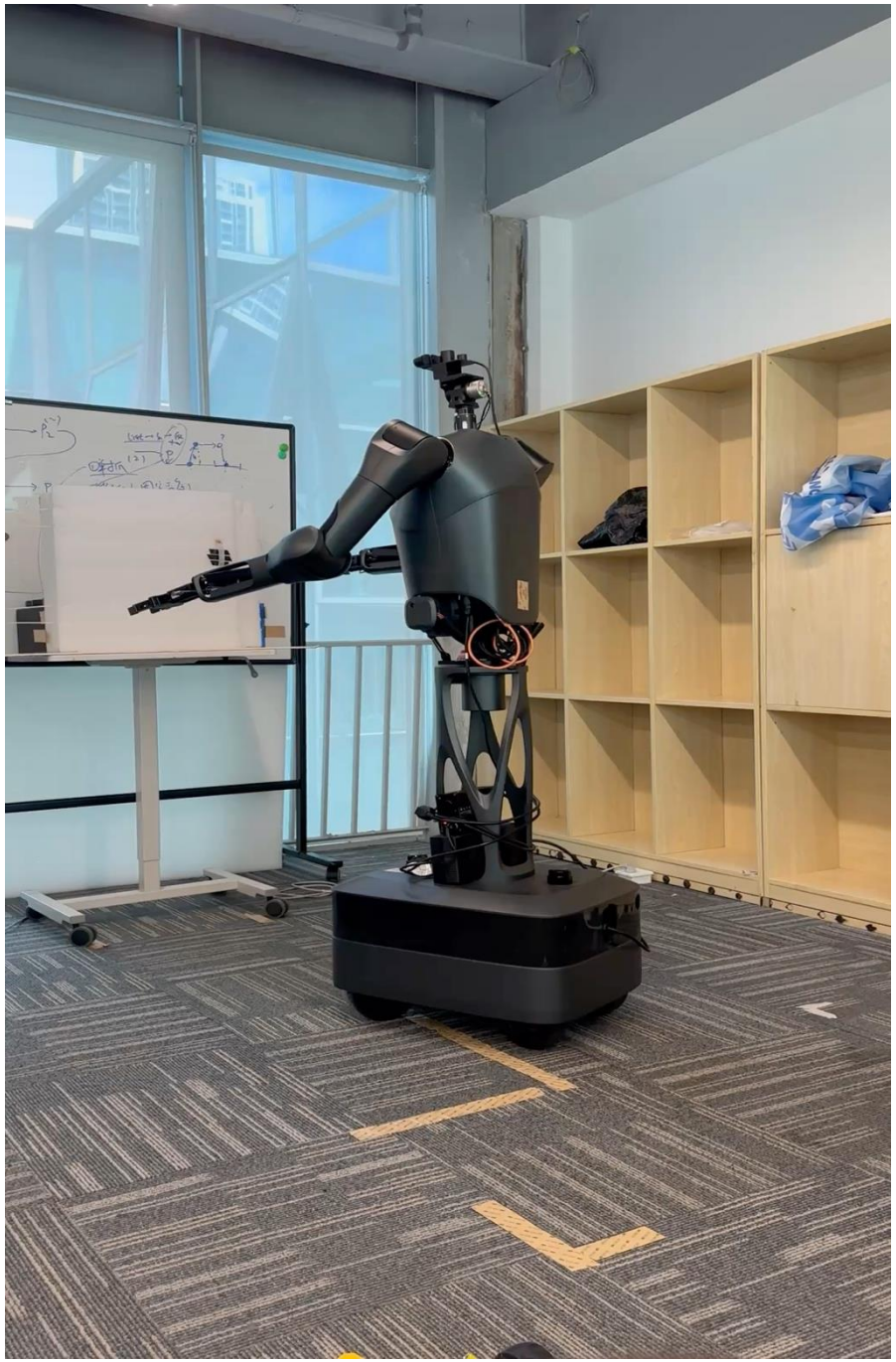


Data collection toward automation



Background
Semi-Humanoid Robot Teleoperation

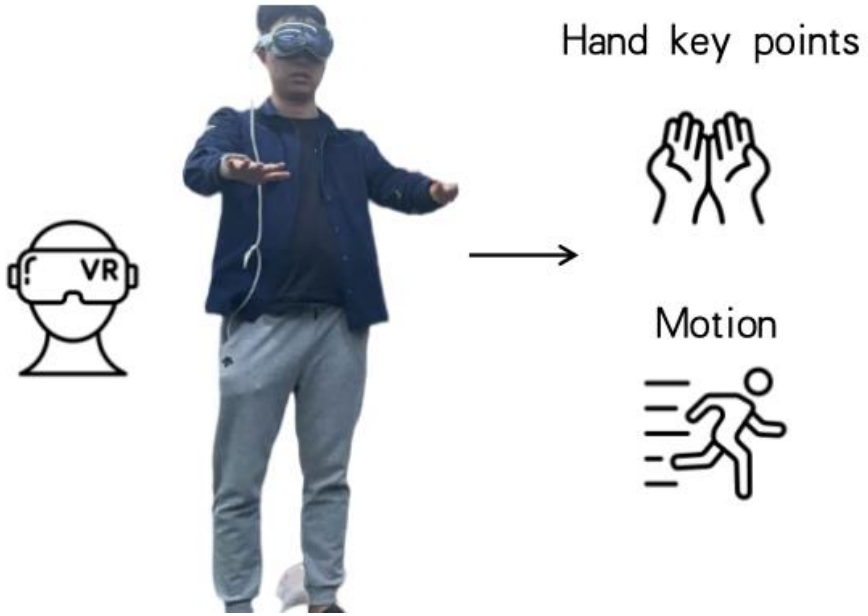
- **Semi-Humanoid** have demonstrated significant potential for application in various environments
- **Semi-Humanoid** will have tremendous potential in **unstructured or semi-structured** engineering environments such as construction



Challenges

Teleoperation of Semi-Humanoid, especially for tasks that require fine manipulation and precise coordination of movements, remains a challenge.

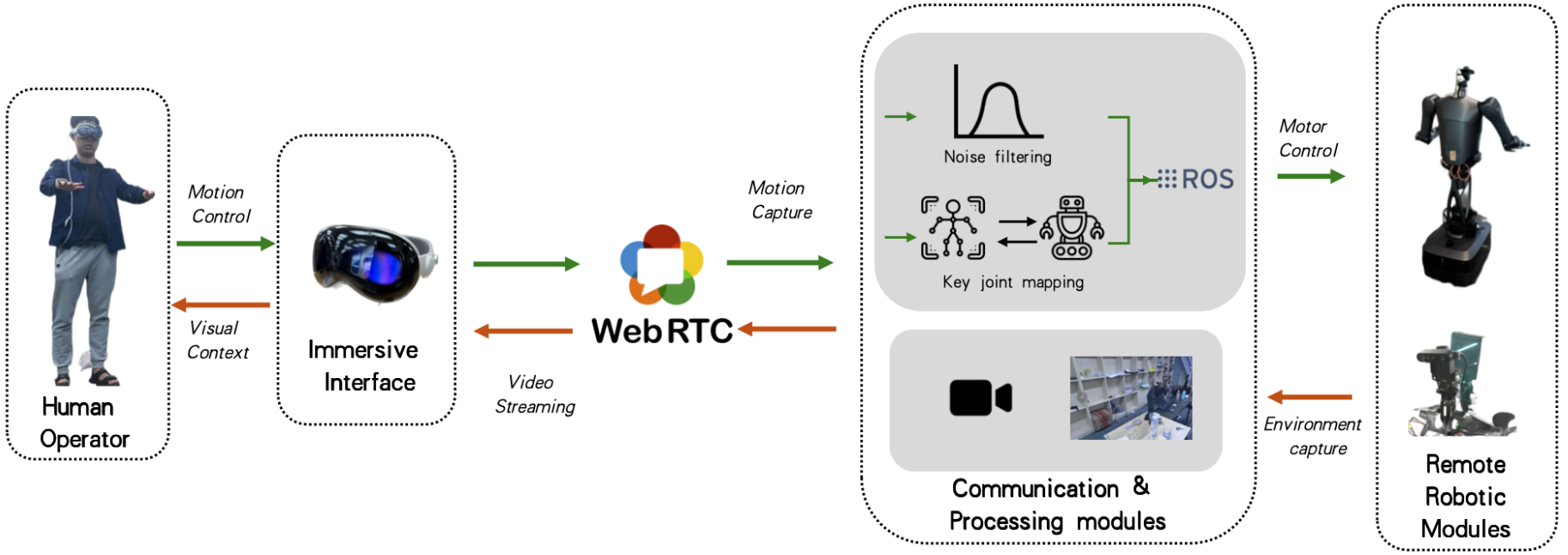
- Noisy and Unstable data.
- Limited degrees of freedom (DoF) of robotic devices pose mapping human motion to robots.



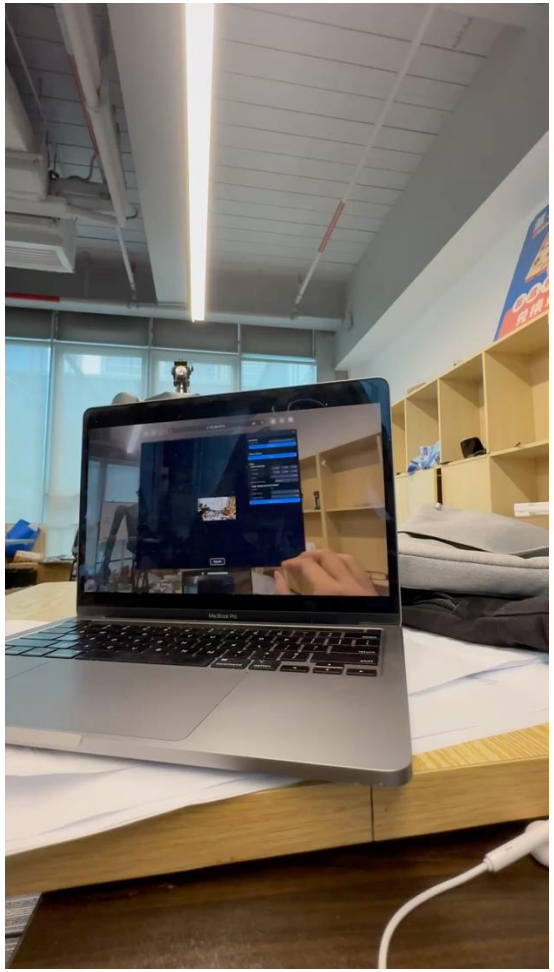
Objective

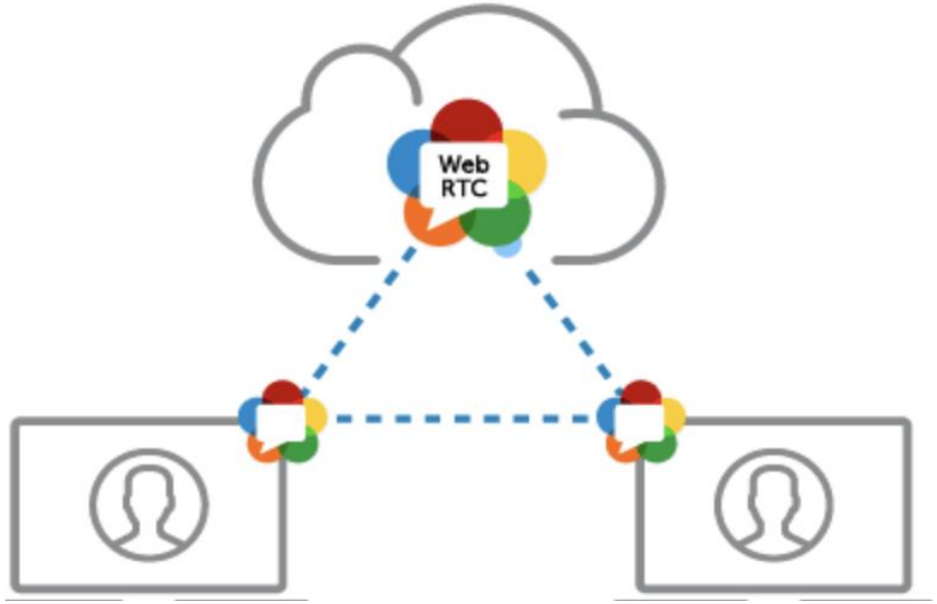
A Teleoperation System that **Filters, Adapts, and Maps noisy and Asymmetrical data collected** from human operators to control a mobile manipulator for construction tasks is needed.

Overall architecture



AVP Real-time Image Display



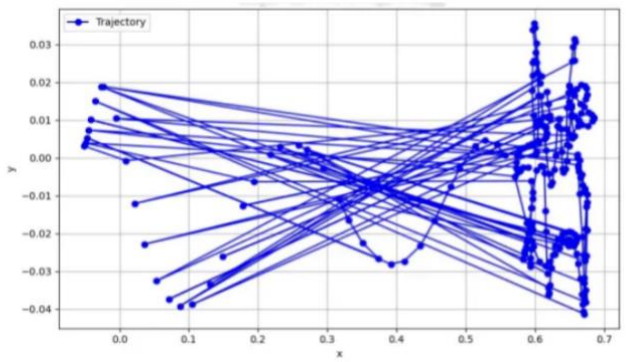


Use WebRTC to establish communication between the robot's head camera and the AVP system,.

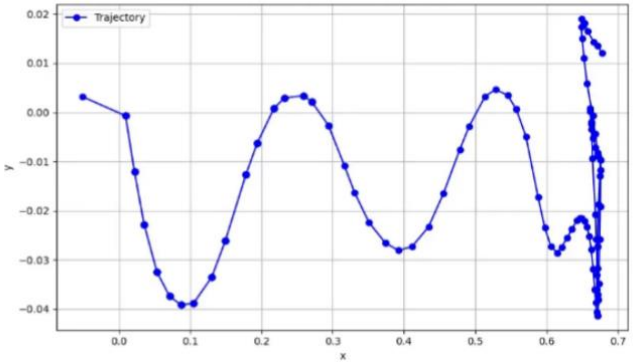
Sensor noise filtering



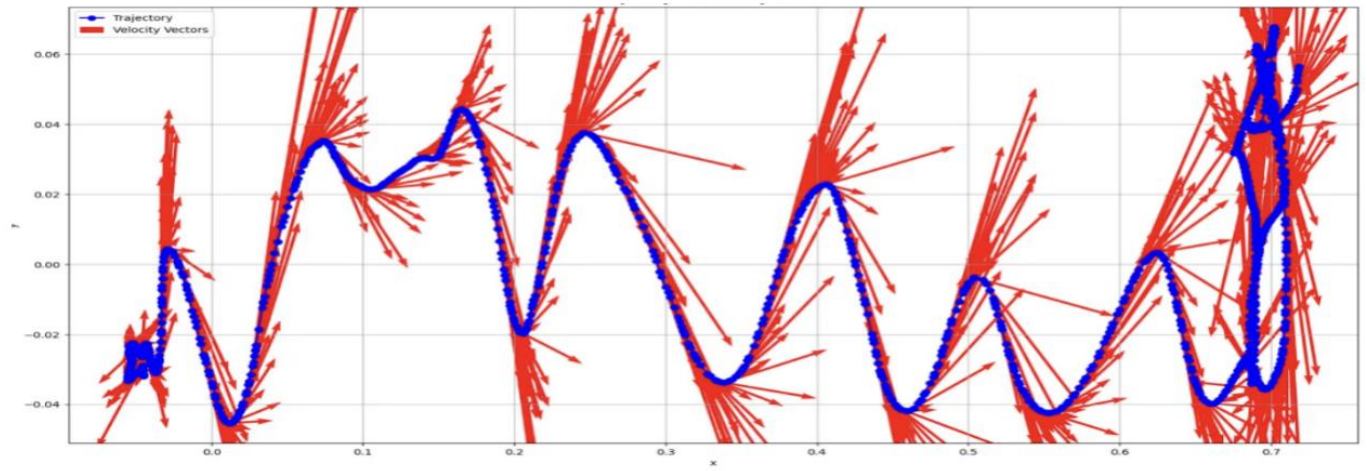
Trajectory with Nosie



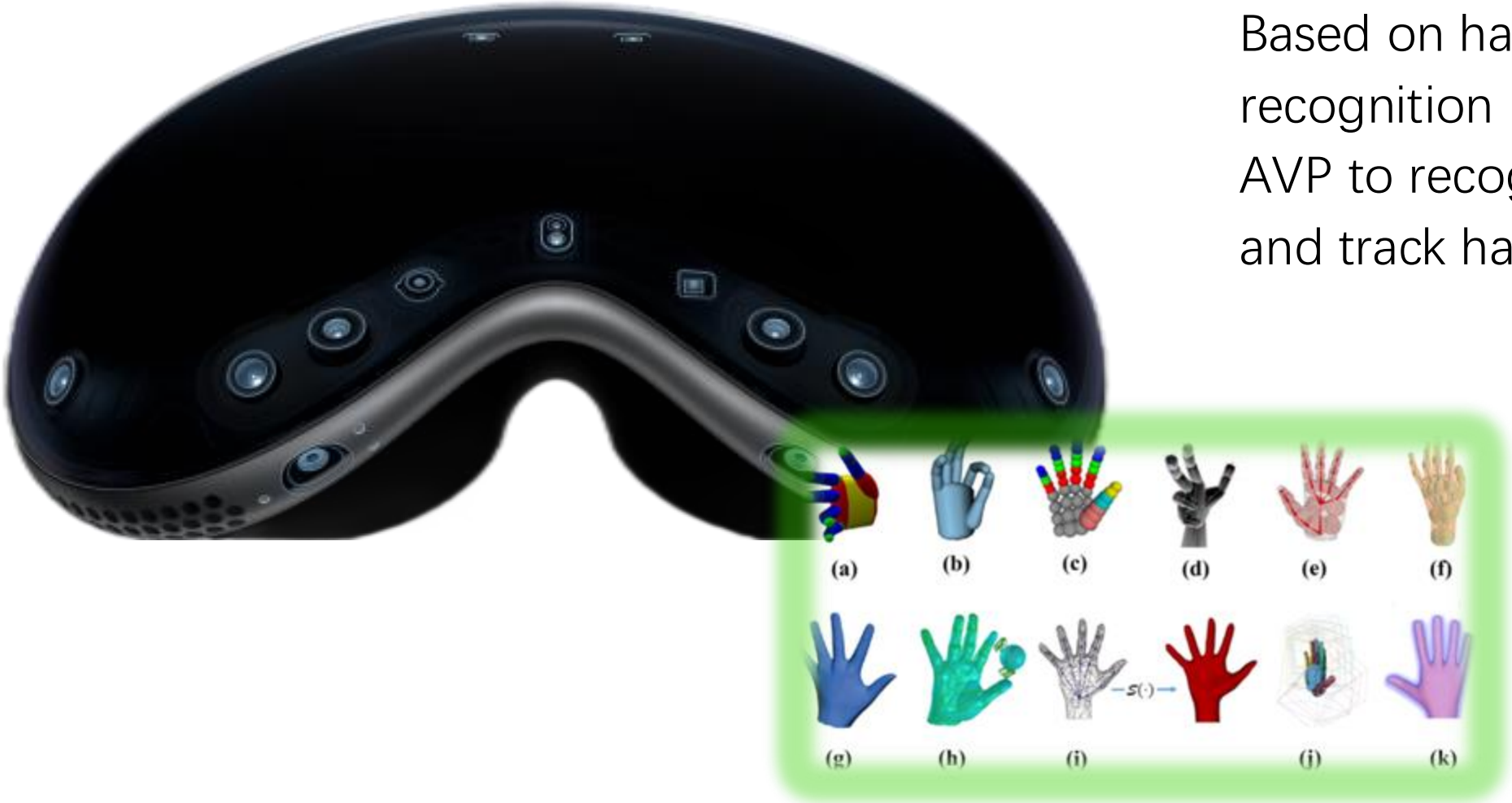
Trajectory with Denosing



Trajectory with Velocity Vectors

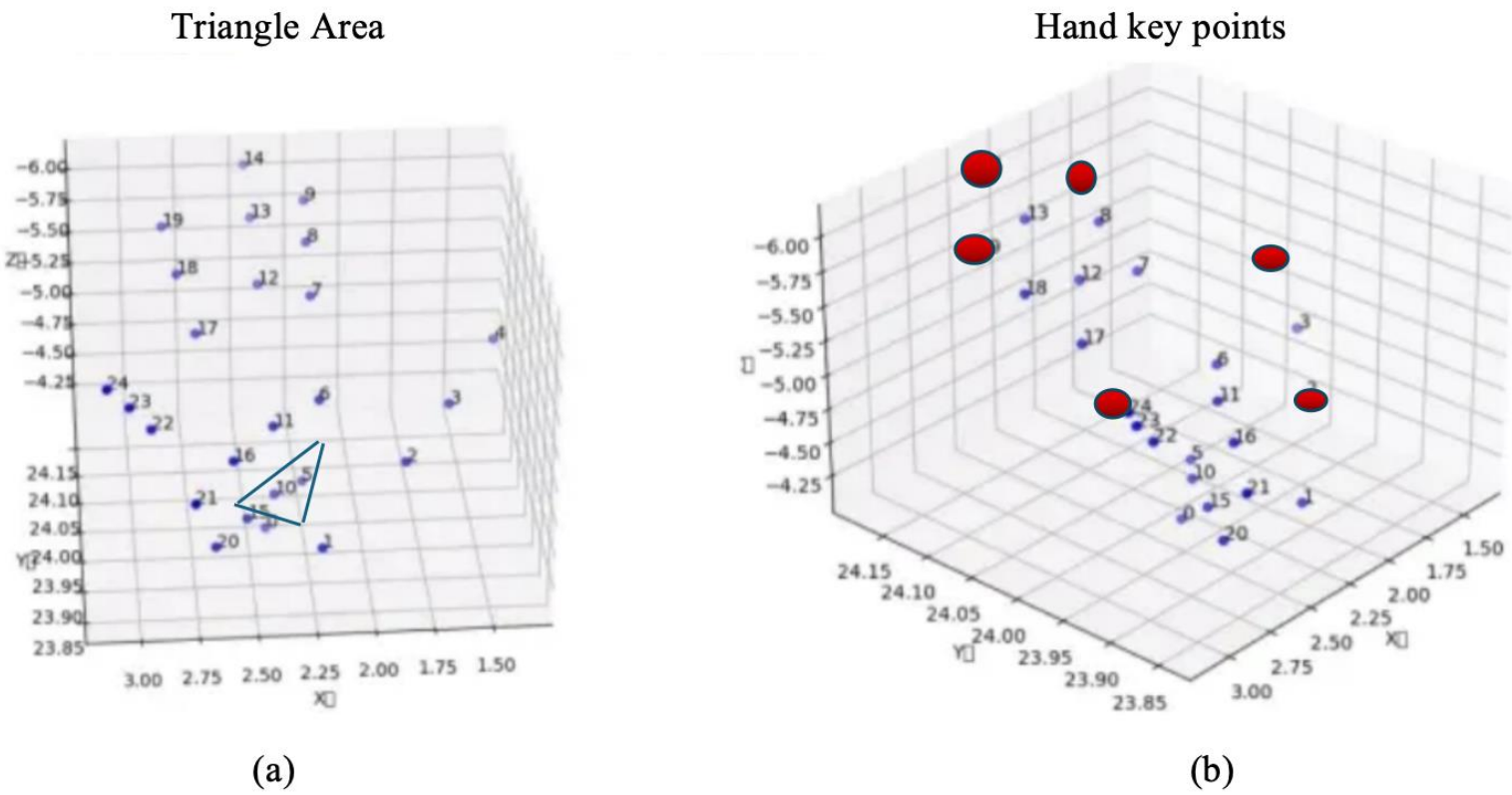


Methodology



Based on hand gesture recognition models, we use AVP to recognize gestures and track hand poses.

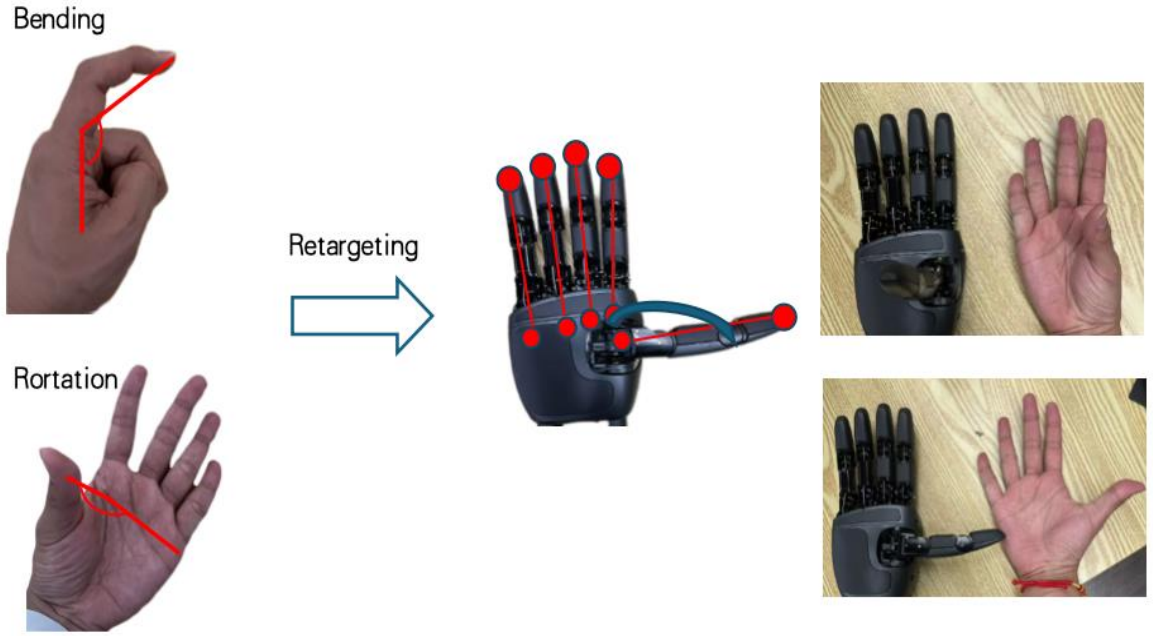
Methodology



Use of **the centroid of a triangle**, formed by tracking three key points, as the palm's central reference point to calculate the control parameters.

Mapping the human hand's six degrees of freedom (five bending motions and one rotation).

Methodology



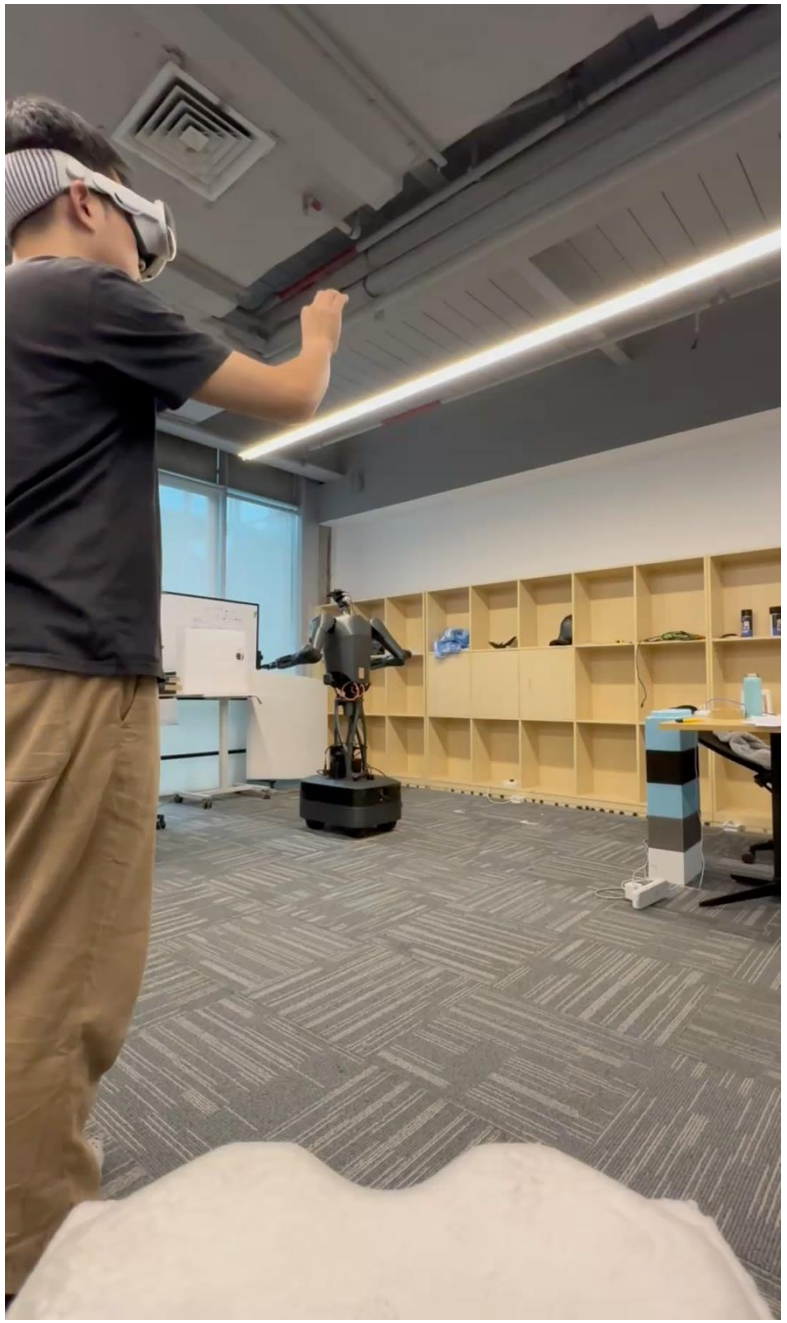
Measuring the angle changes between the **thumb pad and the palm**, we can accurately control the rotation of the thumb.

Construct a triangle area on the tracked human palm using three key points and **calculate the centroid point** as the palm location for thumb rotation calculation.



Experiment

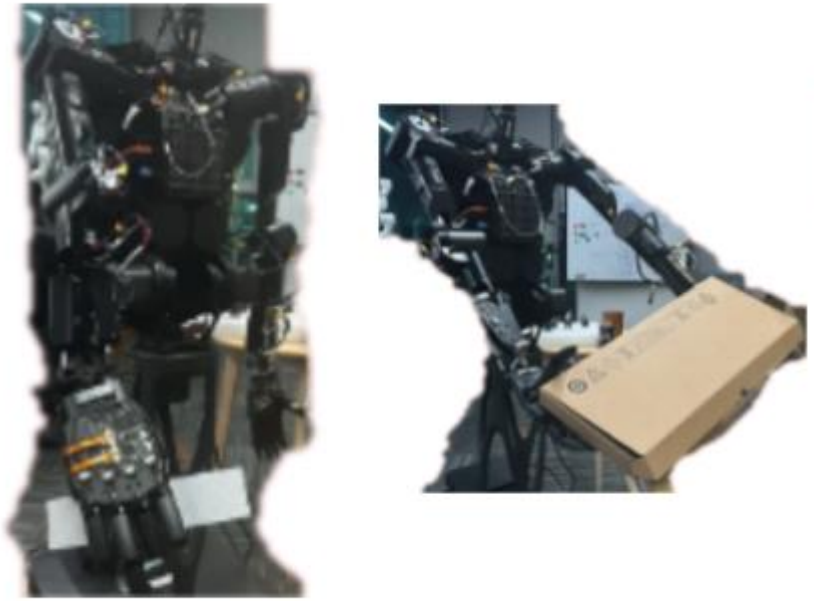
In our experiments, we explored mapping different hand gestures to corresponding operational commands.



Experiment

Findings & Conclusion









References

- Darvish, K., Penco, L., Ramos, J., Cisneros, R., Pratt, J., Yoshida, E., Ivaldi, S., and Pucci, D. (2023). "Teleoperation of humanoid robots: A survey." IEEE Transactions on Robotics, 39(3), 1706-1727.
- Hamner, B., Koterba, S., Shi, J., Simmons, R., and Singh, S. (2010). "An autonomous mobile manipulator for assembly tasks." Autonomous Robots, 28, 131-149.
- Holland, J., Kingston, L., McCarthy, C., Armstrong, E., O'Dwyer, P., Merz, F., and McConnell, M. (2021). "Service robots in the healthcare sector." Robotics, 10(1), 47.
- Li, R., Wang, H., and Liu, Z. (2021). "Survey on mapping human hand motion to robotic hands for teleoperation." IEEE Transactions on Circuits and Systems for Video Technology, 32(5), 2647-2665.
- Song, C. S., and Kim, Y.-K. (2022). "The role of the human-robot interaction in consumers' acceptance of humanoid retail service robots." Journal of Business Research, 146, 489-503.
- Ye, Y., Xia, P., Xu, F., and Du, J. (2023). "Enhance Kinesthetic Experience in Welding Motor Training with Virtual Reality and Robotic Arms." Transactions on Human-Machine Systems.
- Moniruzzaman, M. D., Rassau, A., Chai, D., & Islam, S. M. S. (2022). Teleoperation methods and enhancement techniques for mobile robots: A comprehensive survey. Robotics and Autonomous Systems, 150, 103973.

IEEE 4th International Conference on Intelligent Reality (ICIR 2025)

November 16-18 | Boston, MA, USA



IEEE DigitalReality

WNE GOLDEN BEARS

Northeastern University

Conference Overview

The IEEE 4th International Conference on Intelligent Reality (ICIR 2025) will be held at ***The Cabral Center***, Northeastern University, Boston, Massachusetts, USA, on November 16-18, 2025. The conference committee invites submissions of innovative work on Intelligent Reality from researchers, educators, and industrial engineers. Western New England University and Northeastern University will support this conference.

Contact: xiao.h1@northeastern.edu